

ReSepNet: A Unified-Light Model for Recursive Speech Separation with Unknown Speaker Count

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Abstract

Single-channel speech separation remains a significant challenge, particularly when the number of concurrent speakers is unknown. Existing methods often rely on prior assumptions about speaker count, limiting their real-world applicability. This paper introduces ReSepNet (Recursive Separation Network), a novel, unified model for speaker-independent speech separation that dynamically adapts to an unknown number of speakers. ReSepNet employs a recursive separation architecture, enabling it to iteratively isolate individual voices without prior knowledge of the speaker count. To enhance efficiency and reduce model complexity, we introduce a novel objective function and workflow. We demonstrate the effectiveness of ReSepNet on the WSJ0 datasets, achieving state-of-the-art separation performance and accurate speaker count estimation. Furthermore, ReSepNet generalizes well to mixtures containing four and five speakers, showcasing its robustness and adaptability to challenging scenarios.

Index Terms: Speech separation, recursive, unknown speaker count, improved transformer, single channel separation

1. Introduction

Speech separation, a core challenge in audio processing, aims to isolate individual speech sources from their mixtures. Applications span speech recognition, voice assistants, and speaker verification. Traditional methods like Independent Component Analysis (ICA)[1] and Independent Vector Analysis (IVA)[2] excel in overdetermined scenarios, while underdetermined mixtures often rely on clustering techniques such as those in [3] and Gaussian Mixture Models (GMM) [4]. Deep Neural Networks (DNNs) have revolutionized underdetermined speech separation, with models like Conv-TasNet-gLN [5] achieving state-of-the-art results. However, most existing methods necessitate prior knowledge of the number of speakers in the mixture, a crucial piece of information for model input.

Existing approaches for underdetermined speech separation with unknown speaker counts often exhibit limitations. For instance, [6] employs a Count-Head block to estimate the speaker count and then utilizes a dedicated decoder for the estimated number, restricting the maximum separable speaker count. Alternatively, [7] trains separate models for each possible speaker count and selects the model with the highest output channel activity. This approach also limits the maximum separable count and necessitates a complex multi-model architecture.

Novel approaches in [8, 9] overcome the limitation of training separate models for each speaker count by iteratively separating one source at a time. However, these methods are constrained to scenarios with equal or fewer speakers than those encountered during training and operate in the time-frequency

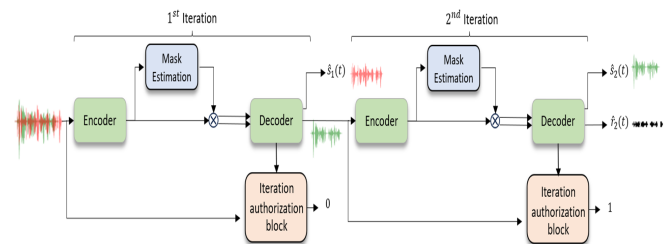


Figure 1: Workflow and block diagram of ReSepNet for separating a mixture of two speakers.

domain. In contrast, recursive approaches in [10, 11] exhibit greater flexibility, enabling separation of mixtures with more speakers than those seen during training. [10] employs iterative separation, while [11] refines the initial separation by re-separating the extracted sources from the original mixture. Despite their potential, these methods suffer from significant drawbacks. The approaches in [8, 9] encounter a heavy model due to the utilization of AlexNet [12] for iteration termination, leading to increased computational complexity. Furthermore, the employment of CNNs in [10] can result in limited receptive fields, while the use of RNNs in [11] necessitates information propagation through numerous intermediate states [13], potentially hindering performance.

ReSepNet, our proposed framework, employs a recursive architecture. At each iteration, it generates two outputs: the separated speech of one speaker and the residual signal containing the remaining speakers. This residual signal then serves as input for the subsequent iteration. To effectively terminate the separation process, we introduce an Iteration Authorization Block (Fig. 1). To enhance separation quality and minimize model complexity, we incorporate several key innovations:

1. **Transformer-based Architecture:** We leverage the strengths of improved Transformer architectures [13, 14], enabling direct context-aware modeling of speech sequences and overcoming the limitations of RNNs and CNNs in capturing long-range dependencies.
2. **Dual-Path structure:** To effectively handle long input sequences, we adopt a Dual-Path structure [15], enabling more efficient analysis of extended audio segments.
3. **Counting method:** We introduce a novel source counting mechanism that effectively overcomes the limitations of existing methods, such as delays and excessive parameter requirements, by leveraging the model's output characteristics.

2. SYSTEM OVERVIEW

ReSepNet operates in the time domain and comprises four key stages: an Encoder, a Mask Estimation module, a Decoder, and an Iteration Authorization Block (Fig. 1). The separation process proceeds iteratively. In each iteration, the Encoder generates an embedding of the input mixture. Subsequently, the Mask Estimation module produces separation masks that isolate the target speaker's contribution. The Decoder then reconstructs the time-domain signal corresponding to the separated speaker. Finally, the Iteration Authorization Block evaluates the residual signal to determine if additional speakers remain. The separation process continues iteratively until the residual signal primarily consists of noise, indicating the completion of the separation process.

In this work, similar to [5, 6, 10] the problem of single-channel speech separation is estimating the sources $s_1(t), \dots, s_c(t) \in R^{(1 \times T)}$, with duration T , from the mixture $x(t) \in R^{(1 \times T)}$ which is defined as (1),

$$x(t) = \sum_{i=1}^C s_i(t) \quad (1)$$

The method separates one source at a time from the input mixture, producing two outputs: the estimated sound of one source, $s_k(t)$ and the residual signal that is a mixture of the remaining sources, $\hat{r}_k(t)$. To separate another source, the remaining mixture is fed back into the model. The operation of the model in the k -th iteration is described in (2).

$$\hat{s}_k(t), \hat{r}_k(t) = F(\hat{r}_{(k-1)}) \quad , \quad \hat{r}_0(t) = x(t) \quad (2)$$

In fact, F is applied recursively on the remaining mixture to separate the remaining sources. Therefore, in order to separate C sources from the mixture, the model have to be used C times.

2.1. Encoder

The Encoder comprises a one-dimensional convolutional network with ReLU activation. This network employs convolutional filters with kernel size L and N output channels. The input signal is processed with a 50% overlap, resulting in a tensor of dimensions $D^{N \times E}$, where E represents the number of output features. This process, which can be mathematically expressed as in (3), effectively extracts informative features from the input mixture x by applying convolutional filters with weights W .

$$X = \text{ReLU}(x * W) \quad (3)$$

2.2. Mask Estimation

As illustrated in Fig. 2, the Mask Estimation module generates separation masks to isolate individual speakers from the input mixture. To effectively handle long-duration mixtures, the Encoder's output is segmented into overlapping chunks of length K , with a 50% overlap. These segments are then organized into a three-dimensional tensor of dimensions $D^{N \times K \times P}$, where P represents the number of segments. This tensor subsequently undergoes Intra and Inter-chunk processing.

2.3. Intra and Inter-chunk processing

The Intra and Inter-chunk processing section, designed to capture both short-term and long-term dependencies, employs a cascading structure consisting of B blocks. This section leverages the Dual-path approach proposed in [15] to effectively

model these dependencies. In the Intra-processing mode, segments of the input tensor with dimensions $D^{N \times K}$ are processed by the Improved Transformer (IT) block [13]. This block performs local processing on the input tensor, as mathematically described in (4).

$$D_b^{\text{Intra}} = IT_b[D_{b-1}^{\text{Intra}}[:, :, z], z = 1, \dots, P] \quad (4)$$

where, $b \in B$ and if $b = 1$, the input to IT is segmented Encoder output.

In the Inter-processing mode, segments of the input tensor with dimensions $D^{N \times P}$ are processed by the IT block. This block performs global processing on the input tensor, capturing long-range dependencies across all segments. The mathematical representation of this process is as (5),

$$D_b^{\text{Inter}} = IT_b[D_b^{\text{Intra}}[:, y, :], y = 1, \dots, K] \quad (5)$$

As stated in (5), the output of local processing enters the global processing mode, which enables the model to better handle long-term inputs.

2.4. Decoder

Following mask generation, the separation masks are applied element-wise to the Encoder's output, resulting in the separated speaker's signal and the residual signal. Subsequently, a transpose convolutional network [16] with kernel size L and N input channels is employed in the Decoder to reconstruct the time-domain representation of the separated speaker's signal from the masked features, as described in (6).

$$s_i(t) = Y_i * V, \quad i = 0, 1 \quad (6)$$

where Y_i represents the masked arrangements for the audio of one speakers ($i = 0$) and the residual signal ($i = 1$), and $Y \in R^{2 \times N}$. In addition, $V \in R^{N \times L}$ represents the parameters of the transposed convolution module.

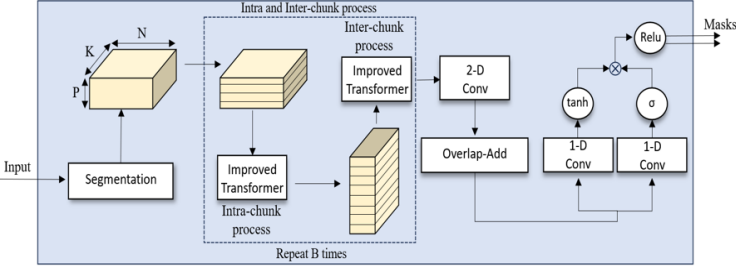
3. Iteration Authorization Block

The Iteration Authorization Block determines whether the residual signal necessitates further processing to extract additional speakers. This block enables the model to be iteratively reused by treating the residual signal as a new input mixture, provided it exhibits characteristics indicative of residual speech. The block's design is informed by an analysis of the model's output behavior. For a mixture of C speakers, the model should ideally execute C iterations. As illustrated in Fig. 2 for a two-speaker mixture, the residual signal in the final iteration exhibits a high degree of noise, while the extracted speech signal closely resembles the original input. These observations guide the design of the Iteration Authorization Block, which incorporates two key components:

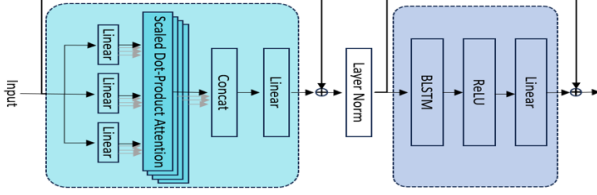
- Cross correlation between the mixed input and the separated source audio.
- Cross correlation between the mixed input and the residual signal.

The Iteration Authorization Block produces a binary output. If the absolute value of any of the calculated cross-correlations exceeds or equals 0.93, the block signals the end of the iteration process by outputting 1. Otherwise, it outputs 0, indicating that further iterations are required.

A. Mask Estimation block



B. Improved Transformer



C. Out put of two speaker mixture

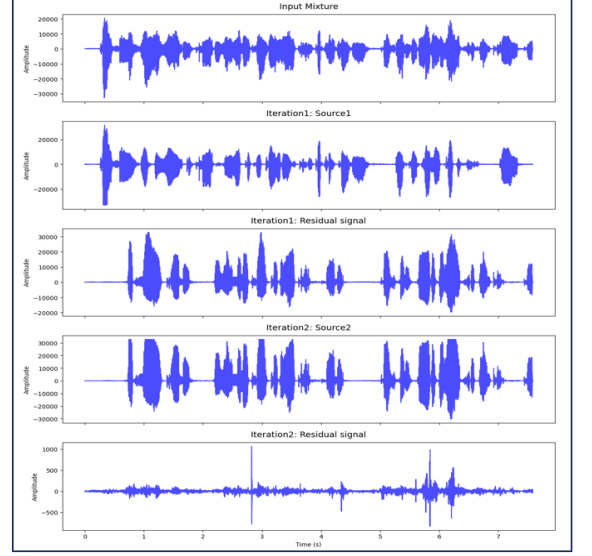


Figure 2: (A) Structure of the Mask Estimation Section, (B) Structure of the Iteration Authorization Block and (C) Separation of a 2 speaker mixture over 2 iterations.

4. Model Training Objective Function

To prevent the model from exhibiting a bias towards a specific number of speakers, the training process must incorporate mixtures containing a diverse range of speaker counts. Consequently, the objective function for model training is defined as in (7).

$$L = \sum_{k=1}^{C-1} \ell_k \quad (7)$$

where, k is the iteration number, C is number of speakers in the input mixture, and ℓ_k is the loss function in the k -th iteration. Since our goal is to increase SI-SNR and solve the permutation problem, we use uPIT [18] and OR-PIT [10] to define ℓ_k , which is denoted as (8):

$$\ell_k = \frac{\max_i l(\hat{s}(t), s_i(t)) + \frac{1}{C-k} l(\hat{r}(t), \sum_{n \neq i} s_n(t))}{C_M - C + k} \quad (8)$$

where, C_M denotes the maximum number of speakers in the model's training mixtures; l is the function, that calculates the Signal-to-Interference-and-Noise Ratio (SI-SNR) for two model's outputs, and $i \in \{1, \dots, C\}$ corresponds to speaker indices. Specifically, the SI-SNR is first calculated between the estimated source signal, $\hat{s}(t)$, and the reference sounds, determining the source with the highest SI-SNR to be $s_i(t)$. Next, the SI-SNR is computed between the residual signal output and the sum of the reference sounds (excluding the reference $s_i(t)$) according to (8). The method for calculating SI-SNR through function l is in [19, 10, 15], and denoted as in (9):

$$l_{\text{SI-SNR}}(\hat{s}(t), s(t)) := 10 \log_{10} \frac{|s_{\text{target}}|^2}{|e_{\text{noise}}|^2} \quad (9)$$

where, $s_{\text{target}} := \frac{\langle \hat{s}(t), s(t) \rangle s(t)}{|s(t)|^2}$ and $e_{\text{noise}} := \hat{s}(t) - s_{\text{target}}(t)$. Here, $\hat{s}(t)$ and $s(t)$ are mean-normalized in the time domain. Normalizing the mean of the sources ensures the scale invariance of the objective function [10].

Unlike uPIT, where the number of permutations for a mixture with C speakers is $C!$, the proposed approach reduces the number of permutations to C , significantly reducing computational complexity. Furthermore, the sum of the remaining sources, $\hat{r}_k(t)$, is consistently trained to appear in the second output channel. This constraint ensures that no trace of the extracted speaker's speech is present in the residual signal, effectively enforcing a separation constraint. By explicitly attending to the separated speaker in one output channel and the residual signal in the other, the model's separation performance is significantly enhanced, as demonstrated in [10].

5. Model Training

Model training was conducted using the WSJ0 dataset [20], following the methodologies outlined in [21, 10]. Audio recordings from distinct speakers were randomly mixed with Signal-to-Noise Ratios (SNRs) ranging from -2.5 dB to 2.5 dB, resulting in the creation of the WSJ0-2mix and WSJ0-3mix datasets for two- and three-speaker scenarios, respectively. ReSepNet was exclusively trained on a combined dataset of WSJ0-2mix and WSJ0-3mix, setting the maximum number of speakers during training C_M to 3. To reduce computational demands, audio samples were downsampled to 8 kHz. For training, the audio data was segmented into 4-second windows. Segments shorter than 2 seconds were discarded, while shorter segments were zero-padded to achieve a consistent 4-second duration. To evaluate the model's generalization capabilities, additional WSJ0-4mix and WSJ0-5mix datasets were created for four- and five-speaker test scenarios.

Train epoch is 83 (due to numerical instability in later epochs, resulting in NaN outputs), and the Adam [22] optimizer is used with gradient clipping 5. The learning rate (lr) is initially increased and then reduced by multiplying by 0.98 every two

Table 1: Comparison of model performance on WSJ0 datasets. The table presents the **SI-SNRI (dB)** and the number of parameters for models on WSJ0-2mix, WSJ0-3mix, WSJ0-4mix, and WSJ0-5mix test datasets. Symbols used: '*' indicates a model trained for a fixed number of speakers, '-' denotes unavailable data, and '+' signifies that the number of parameters exceeds the stated value.

Model	WSJ0-2mix	WSJ0-3mix	WSJ0-4mix	WSJ0-5mix	# Param
Conv-TasNet-gLN* [5]	14.6	11.6	-	-	8.8M
DPRNN* [15]	18.8	-	-	-	2.6M
DPTNet* [13]	20.2	-	-	-	2.69M
Sepformer* [17]	20.4	17.6	-	-	26M
MULCAT [7]	20.12	16.70	12.82	10.50	7.5M
OR-PIT [10]	14.8	12.6	10.2	-	+60M
DPRNN + OR-PIT [11]	17.8	13.7	9.9	6.2	+60M
Coarse-to-fine [11]	19.8	15.4	11.4	7.3	+60M
ReSepNet (Ours)	21.16	19.19	14.91	12.03	2.8M

epochs.

$$\text{lr} = \begin{cases} k_1 \cdot d_{\text{model}}^{-0.5} \cdot n \cdot \text{warmup}_n^{-1.5} & , n \leq \text{warmup}_n \\ k_2 \cdot 0.98^{\text{Epoch}/2} & , n > \text{warmup}_n \end{cases} \quad (10)$$

where, n is the step number, the value of d_{model} is 64, and K_1 and K_2 are tunable numerical values being set during model training, where K_1 is set to 0.2, K_2 to $4e^{-4}$, warmup_n is set to 4000, and the batch size is set to 1, similar to [15, 13, 14].

6. Experimental Results

The model was trained on a server with a high-end NVIDIA GeForce RTX 3090 GPU with 24 GBytes RAM, and every epoch of training required approximately 6.2 hours. Remarkably, despite being trained exclusively on WSJ0-2mix and WSJ0-3mix datasets, ReSepNet successfully generalizes to separate challenging WSJ0-4mix and WSJ0-5mix mixtures, demonstrating its robustness and adaptability. This achievement is particularly noteworthy considering the significantly reduced model complexity compared to existing state-of-the-art methods. Separation samples are available on our GitHub repository for independent evaluation and analysis¹. Furthermore, ReSepNet surpasses the performance of models designed for a known number of speakers, such as DPTNet [13] and SepFormer [17], which utilize the Dual-path method without the dynamic mixing capability of our approach.

To rigorously assess the accuracy of the Iteration Authorization Block, we conducted extensive evaluations on a dedicated test set comprising 3000 samples for each of the 1, 2, and 3 speaker conditions, drawn from the WSJ0 test sets, WSJ0-2mix, and WSJ0-3mix, respectively. A successful classification is defined as the block accurately outputting zeros for all but the final iteration, where a 1 is expected. Our proposed approach achieves an impressive accuracy of 96.6%, surpassing the performance of existing recursive models [10, 11] which exhibit an accuracy of 95.7%. Notably, unlike these models that rely on computationally expensive AlexNet for iteration control and require 10-second input segments of mel spectrograms, our novel Iteration Authorization Block solely leverages cross-correlation analysis, eliminating the need for feature extraction and imposing no constraints on input segment length.

¹<https://hadializadeh1.github.io/ReSepNet/>

Furthermore, the Real-Time Factor (RTF) computed on WSJ0-2Mix for single iteration and two iterations demonstrates that our proposed recursive method remains efficient while outperforming baseline approaches [5, 13, 15].

7. Conclusion

This paper presented ReSepNet, a novel recursive speech separation framework that dynamically adapts to mixtures with an unknown number of speakers. By integrating a transformer-based architecture with a recursive separation workflow and a lightweight Iteration Authorization Block, ReSepNet eliminates the need for prior assumptions about speaker counts while achieving state-of-the-art performance. The key innovations including a dual-path transformer structure for efficient long-context modeling, a permutation-invariant training objective that reduces computational complexity, and a cross-correlation-based iteration termination mechanism achieves 96.6% accuracy in speaker count estimation. Experimental results on WSJ0-2mix and WSJ0-3mix datasets demonstrate ReSepNet's superiority over existing methods, with SI-SNRI improvements. This work opens new directions for real-world speech separation systems, particularly in scenarios with unpredictable speaker counts. Future research will explore extending ReSepNet to real-time applications and integrating speaker diarization for fully end-to-end solutions.

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